

SQL Interview Question: Delete Duplicate Rows

Scenario

An **employees** table contains duplicate records due to multiple data imports. You need to remove duplicate rows while keeping only one record.

Sample Table

emp_id	emp_name	email	salary
1	Amit	amit@gmail.com	50000
2	Rahul	rahul@gmail.com	60000
3	Amit	amit@gmail.com	50000
4	Neha	neha@gmail.com	70000
5	Rahul	rahul@gmail.com	60000

Solution 1: Using **ROW_NUMBER()** (Recommended for PostgreSQL)

```
WITH duplicates AS (  
    SELECT *,  
           ROW_NUMBER() OVER (  
               PARTITION BY email  
               ORDER BY emp_id  
           ) AS rn  
    FROM employees  
)  
DELETE FROM employees  
WHERE emp_id IN (  
    SELECT emp_id  
    FROM duplicates  
    WHERE rn > 1  
);
```

Explanation

- **PARTITION BY email** groups duplicate emails.
- **ROW_NUMBER()** assigns:
 - First record → **rn = 1** (kept)

- Remaining records → `rn > 1` (deleted)

Solution 2: Delete Exact Duplicate Rows (Multiple Columns)

If duplicates are identified by `emp_name`, `email`, and `salary`:

```
WITH duplicates AS (  
    SELECT *,  
           ROW_NUMBER() OVER (  
               PARTITION BY emp_name, email, salary  
               ORDER BY emp_id  
           ) AS rn  
    FROM employees  
)  
DELETE FROM employees  
WHERE emp_id IN (  
    SELECT emp_id  
    FROM duplicates  
    WHERE rn > 1  
);
```

Output After Deletion

emp_id	emp_name	email	salary
1	Amit	amit@gmail.com	50000
2	Rahul	rahul@gmail.com	60000
4	Neha	neha@gmail.com	70000

Best Practices

- Always preview duplicates before deleting:

```
SELECT *,  
       ROW_NUMBER() OVER (  
           PARTITION BY email  
           ORDER BY emp_id
```

```
) AS rn  
FROM employees;
```

- Take a backup before running **DELETE**.
- Use a **primary key** (like **emp_id**) in the **ORDER BY** clause to decide which record to keep.

Interview Follow-up Questions

Q1. Why use **ROW_NUMBER()** instead of **RANK()**?

- **ROW_NUMBER()** gives a unique sequence (1, 2, 3...), making it easy to keep exactly one row.
- **RANK()** and **DENSE_RANK()** can assign the same rank to multiple rows, so they're not suitable for deleting duplicates.

Q2. How do you keep the latest record instead of the oldest?

```
ROW_NUMBER() OVER (  
    PARTITION BY email  
    ORDER BY created_date DESC  
)
```

This keeps the most recently created record and deletes the older duplicates.